



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
RAMAPURAM CAMPUS
FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC YEAR (2022-2023)
CONTINUOUS LEARNING ASSESSMENT-1



Sub Code/Name : 18CSE390T COMPUTER VISION Set: EVEN
Class : III Year / V Sem / B.Tech (AIML)
Date : 13-09-2022
Max Marks : 25 Duration : 60 mins

PART A (5x1= 5)

ANSWER ALL THE QUESTIONS

Q.No.	Question	Marks	CO	BL	PI
1.	A translation is applied to an object by _____. a) Enlarging the Object b) Repositioning it along with straight line path c) Repositioning it along with circular path d) Shrinking the Object	1	1	1	1.2.2
2.	The two-dimensional translation equation in the matrix form is _____. a) $P' = P + T$ b) $P' = P - T$ c) $P' = P * T$ d) $P' = p$	1	1	1	1.2.1
3.	The rotation axis that is perpendicular to the xy plane and passes through the pivot point is known as _____. a) Rotation b) Translation c) Scaling d) Shearing	1	1	1	1.6.1
4.	In controllable interaction user can change the attributes of the _____. A) Images B) Widgets C) Videos D) Audios	1	1	2	2.5.1
5.	If the direction of the projection is normal then it is called as _____. A) Orthographic parallel projection B) Oblique parallel projection C) Perspective Projection D) Ortho-Oblique Projection	1	1	2	2.6.1

PART-B (2x4= 8)
ANSWER ALL THE QUESTIONS

Q.No.	Question	Marks	CO	BL	PI
6.	Define BRDF. What is Helmholtz reciprocity?	4	1	3	3.6.2
7.	Differentiate between Discrete Fourier Transform and Fast Fourier Transform.	4	1	3	3.6.4

PART-C (1x12= 12)
ANSWER ALL THE QUESTIONS

Q.No.	Question	Marks	CO	BL	PI
8.a	Illustrate briefly about Orthography and Para-perspective in 2D and 3D geometric primitives.	12	1	2	2.6.4
8.b	Explain the following Linear Filtering techniques, ❖ Separable filtering ❖ Band-pass and steerable filters	12	1	1	1.6.1

Sub Code/Name	18CSE390T – Computer Vision	Set	EVEN
Year/Sem/Branch	III/ V/ B.Tech-CSE-AIML A,B,C	Date	17.10.22
Max. Marks	50	Duration	90 Mins.

PART A (10 X 1= 10)

ANSWER ALL THE FOLLOWING QUESTIONS

Q.No.	MCQ Questions	Marks	CO	BL	PI
1.	For edge detection we observe a) intensity transition b) shape transition c) color transition d) sign transition	1	2	1	1.6.1
2	The direction of angle to the gradient is a) Orthogonal b) Isolated c) Isomorphic d) Isotropic	1	2	1	1.6.1
3	Edge detection in images is commonly accomplished by performing a spatial --- of the image field. a) Smoothing Filter b) Integration c) Differentiation d) Min Filter	1	2	2	1.6.1
4	Multi-dimensional hashing maps descriptors into _____ based on some function applied to each descriptor vector. a) fixed size buckets b) variable sized buckets c) table d) Dbms	1	2	2	1.6.1
5	Isolated edge points can also be grouped into _____ a) Pixel b) region c) Longer curves or contours, as well as straight line segments d) Contour	1	2	1	1.6.1
6	Techniques like Livewire or Intelligent Scissors are used in a. Model based segmentation b. Semi automatic segmentation c. Threshold segmentation d. Segmentation	1	3	1	1.6.1

7	Example of Active Contour a. Snakes, intelligent scissors, level set b. Successive Approximation c. Hough Transform d. Scissors	1	3	1	1.6.1
8	An Approach which optimizes the contour in real time as the user is drawing a) Intelligent Scissors System b) Gaussian c) Similarity d) Edge	1	3	1	1.6.1
9	In level set which defines the curve a. Contrast b. Quantization c. Sampling d. Zero crossing of a characteristic function	1	3	1	1.6.1
10	Split and merge technique is a. Image Restoration Technique b. an Image Processing Technique Used To Segment An Image c. Image Enhancement Technique d. Image Acquisition Technique	1	3	1	1.6.1

PART B (4 X 4 = 16)

ANSWER ANY FOUR OUT OF SIX QUESTIONS

Q. No.	Questions	Marks	CO	BL	PI
11	Discuss about Bias and Gain normalization	4	2	1	2.5.1
12	Explain briefly about Vanishing points	4	2	2	2.5.2
13	Write short notes on Edge Linking	4	2	2	2.5.4
14	Discuss in detail about Snakes	4	3	2	2.5.1
15	Difference between Divisive and Agglomerative algorithms in Cluster analysis.	4	3	2	2.6.4
16	Write short note on Pose Estimation.	4	3	2	2.6.2

PART C (2 X 12 = 24)

ANSWER EITHER OF OR IN EACH UNIT

Q. No.	Questions	Marks	CO	BL	PI
17	a) Explain in detail about Feature Detection techniques with relevant examples and diagrams.	12	2	3	2.6.4
	OR b) What are Feature Descriptors? Explain the following Feature Descriptors: ii) SIFT iii) GLOH.	12	2	2	2.7.1
18	a) List the approaches used to locate Boundary Curves in Images. Explain Intelligent Scissors and Level Set in detail.	12	3	1	2.7.1
	OR b) Illustrate the Expectation Maximization algorithm in K-means and Mixture of Gaussians.	12	3	3	2.7.1

CONTINUOUS LEARNING ASSESSMENT-III

Sub Code /Name	18CSE390T – Computer Vision	Set	ODD
Year/Sem/Branch	III/ V/ B.Tech-CSE-AIML	Date	17.11.2022
Max. Marks	50	Duration	90 Mins.

PART A (10 × 1 = 10 marks)
ANSWER ALL THE QUESTIONS

Q.No.	MCQ Questions	Mark	CO	BL	PI
1.	The fundamental matrix is given by _____. a. $F = K_1^T E K_2^{-1}$ b. $F = K_1^{-T} E K_2^{-1}$ c. $F = K_1^T E K_2^{-1}$ d. $F = K_1^{-T} E K_2^{-1}$	1	4	1	1.6.1
2.	Most modern cameras have _____ pixels and an image center near the _____ of the image a. round, center b. rectangle, corner c. square, middle d. round, edge	1	4	1	1.6.1
3.	Which one of the following is the point of intersection of the line joining the camera centers with the image plane? a. Epipole b. Axis c. Point of projection d. Point at infinity	1	4	2	1.6.1
4.	Fourier-based alignment relies on the fact that the Fourier transform of a shifted signal has the same magnitude as the original signal, but a _____ phase	1	4	2	1.6.1

	a. exponentially varying b. linearly varying c. unvarying d. constant				
5.	Structure from motion is a _____ in structure and motion. a. Bipartite problem b. Graph Coloring Problem c. Travelling salesman problem d. Normalized Cut	1	4	1	1.6.1
6.	Triangulation is the converse of _____ problem. a. Direct Linear Transform b. Pose Estimation c. 2D Motion Estimation d. Rigid Transform	1	5	1	1.6.1
7.	For converting a projective reconstruction into a metric one, _____ techniques have been developed. a. Orthographic Projection b. Projection Matrix c. Self-calibration d. Epipolar	1	5	1	1.6.1
8.	The normal vector perpendicular to the line $\hat{n}$ can be expressed as a function of two angles (θ, ϕ) , such that $\hat{n} = (\cos \theta \cos \phi, \sin \theta \cos \phi, \sin \phi)$ using _____. a. Spherical coordinates b. Cylindrical coordinates c. Planar coordinates d. Vector coordinates	1	5	1	1.6.1
9.	Professional panoramic photographers often use pan-tilt heads that make it easy to control the tilt and to stop at specific _____ in the rotation angle. a. error b. bias c. weight d. detents	1	5	1	1.6.1
10.	Radial distortion can be estimated _____. a. ahead of time b. just in time c. using alignment d. by matching pixels	1	5	1	1.6.1

PART B (4 × 4 = 16 marks)
ANSWER ANY 4 QUESTIONS OUT OF 6

Q. No.	Questions	Mark	CO	BL	PI
11.	What is the need for Projective Reconstruction?	4	4	1	2.5.1
12.	Write short notes on the Bundle Adjustment techniques involved in accurately recovering structure and motion.	4	4	2	2.5.2
13.	Give the equation and briefly explain the components of Brightness Constancy Constraint equation.	4	4	2	2.5.2
14.	Explain the Projection from 3D to Cylindrical coordinate	4	5	2	2.5.1
15.	Explain the process of Recognizing Panoramas.	4	5	2	2.5.2
16.	What is parallax? How can it be removed?	4	5	2	2.6.2

PART C (2 × 12 = 24 marks)
ANSWER ALL THE QUESTIONS

Q. No.	Questions	Mark	CO	BL	PI
17.	a) What is Fourier-based Alignment? Explain it in detail with necessary examples.	12	4	3	2.6.4
	OR				
18.	b) Explain Hierarchical Motion Estimation in detail.	12	4	2	2.7.1
	OR				
18.	a) Explain Parametric Motion Models in detail with necessary figures.	12	5	1	2.7.1
	OR				
	b) Explain the concept of Bundle Adjustment in detail.	12	5	3	2.7.1